

HNSS17 · Vascular & Ventricular Systems of the CNS — Chart

Blood–brain barrier, oxygen and nutrient delivery, arteries, vascular compromise, veins and sinuses, and CSF | **STEP 1** = beyond class content ·

EXAM EMPHASIS = flagged in class as testable

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Blood–Brain Barrier & Delivery of Oxygen and Nutrients

The blood–brain barrier

	WHAT	EXAMPLES
Structure	Brain endothelial cells joined by tight junction proteins , mostly without fenestrations	+ basement membrane + astrocyte foot processes STEP 1
Blocks	High-molecular-weight and charged (hydrophilic) substances	Proteins; drugs designed to stay out (Allegra /fexofenadine, many chemo drugs)
Lets through	Small or lipophilic substances	L-dopa, estrogen , O ₂ , CO ₂
Transporters	Hydrophilic needs facilitated transport	Glucose (GLUT1)
Endocytosis	Engage the transferrin receptor → transport by endocytosis	Biologic drug for Hunter syndrome (lysosomal storage of complex polysaccharides; mostly young boys)
Disrupted BBB	Drugs can act where the barrier is broken	Trauma, infarct, tumor, infection

A CNS drug must be **lipophilic** or attached to a sequence that is **actively transported**. A drug made **not** to cross avoids CNS side effects (non-drowsy antihistamines). Parkinson: L-dopa (crosses) + carbidopa (doesn't) **STEP 1**

Delivery of O₂ and glucose

MECHANISM	HOW IT WORKS
Oxygen	Released from hemoglobin in regions of low pH
Fuel	Glucose = main ATP source; can be supplemented by lactate and ketone bodies
Local flow	Vasodilation where ATP is low: areas of high neural activity or hypoxia
Global flow	Autoregulation keeps flow relatively constant across a range of arterial pressures (MAP ~60–150 mmHg STEP 1)
Chronic hypertension	Alters vessel structure → ↑ risk of stroke and dementia

★ **USMLE high-yield**: BBB = **endothelial tight junctions**, **no fenestrations**: big or charged molecules out; small or lipophilic in (L-dopa, estrogen); glucose by GLUT1; biologics by **transferrin-receptor endocytosis**. Circumventricular organs sense blood (**area postrema** → **vomiting**; subfornical organ/OVLT → thirst, fever) or secrete into it (pituitary, **pineal** → **melatonin**). Brain = 2% of weight, **20% of O₂**; unconscious in 10 s, irreversible damage at 6 min. PaCO₂ ↑ → vasodilation → ↑ ICP; hyperventilation lowers ICP **STEP 1**

Circumventricular organs: areas of incomplete barrier

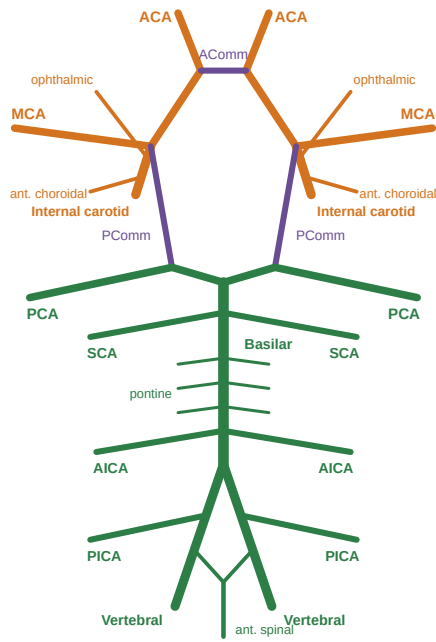
CIRCUMVENTRICULAR ORGAN	DOES	RESULT
Area postrema	Senses blood toxins	Vomiting (chemoreceptor trigger zone STEP 1)
Subfornical organ + organum vasculosum (hypothalamus)	Sense water/ion balance and immune signals	Thirst; fever , warmth-seeking, loss of appetite (feeling sick)
Hypothalamus–pituitary	Releases peptides into blood	Growth, homeostasis, reproduction
Pineal gland	Releases molecules into blood	Melatonin → circadian rhythm

Numbers to know

FACT	NUMBER
Brain's share of body O₂ + calories	20%
Brain's share of body weight	2%
Intracranial neurologic deficits that are vascular	50%
No blood flow → unconscious	~10 seconds
No blood flow → irreparable damage or death	~6 minutes
Survival with mild hypothermia	10–20 minutes , little or no damage

Major Arteries of the CNS & What They Perfuse

Circle of Willis **HIGH YIELD**



■ **Anterior circulation** (internal carotids) · ■ **Posterior circulation** (vertebrals → basilar) · ■ **Communicating** (link the two)

Arteries and their territories

ARTERY	SOURCE	SUPPLIES (LECTURE)	LESION STEP 1
ACA	Internal carotid	Cortex (medial frontal + parietal)	Contralateral leg weakness/sensory loss
MCA	Internal carotid	Cortex (lateral) + striatum (lenticulostrlates)	Contralateral face + arm ; aphasia (dominant) or neglect
PCA	Basilar	Cortex (occipital, inferior temporal) + thalamus	Contralateral homonymous hemianopia, macular sparing
Anterior choroidal	Internal carotid	Hippocampus, amygdala ; choroid plexus	—
Ophthalmic	Internal carotid	Eye	Amaurosis fugax / CRAO
SCA	Basilar	Upper cerebellum	Ataxia
AICA	Basilar	Lower cerebellum, lateral pons, inner ear	Lateral pontine: facial palsy, ipsilateral deafness
PICA	Vertebral	Posteroinferior cerebellum, lateral medulla	Wallenberg : vertigo, dysphagia, hoarseness, crossed pain/temp loss, Horner
Pontine (short circumferential)	Basilar	Pons → pontine infarctions	Basilar occlusion → locked-in
Anterior spinal (ASA)	Both vertebrals (merge)	Anterior cord	Motor + pain/temp loss; dorsal columns spared
Posterior spinal (PSA)	Vertebral or PICA	Posterior cord (stay paired, in grooves)	Dorsal column loss

Spinal cord: ASA + PSA run down the cord off the vertebrals (or PICA) and are **reinforced by radicular arteries** as they descend → localized ischemia possible. **Vertebral arteries:** branch of the **subclavian** → **transverse foramina** of cervical vertebrae → **foramen magnum** → join as the **basilar**. Damaged by **vertebrobasilar occlusion** (atherosclerosis) or **vertebral artery dissection** (trauma, neck distortion). **Watershed areas** = border zones between ACA, MCA and PCA territories (hypotension). **External carotid** → neck, face, scalp.

★ **USMLE high-yield:** **Anterior circulation** = internal carotids → ACA + MCA (+ ophthalmic, anterior choroidal). **Posterior** = vertebrals → basilar → PCA; cerebellum via **SCA, AICA** (basilar) and **PICA** (vertebral); pons via short circumferential branches. **AComm** joins the two ACAs; **PComm** joins the internal carotid to the PCA. Deep structures: **MCA** → **striatum**, **PCA** → **thalamus**, anterior choroidal → hippocampus/amygdala. Most common aneurysm: AComm; PComm aneurysm → CN III palsy **STEP 1**.

CNS Vascular Compromise

Ischemic

embolism · thrombosis

Hemorrhagic

vessel breaks: aneurysm, AVM, dural fistula, hypertension, trauma

Hemorrhage treatment

control BP · reduce ICP · surgically evacuate the hematoma

Types of vascular compromise

PROBLEM	MECHANISM	CLUES	TREATMENT
Cerebral embolism	Traveling clot obstructs a cerebral vessel → ischemia (ischemic stroke)	Sudden deficit; AFib, carotid plaque STEP 1	Reperfusion (tPA ≤4.5 h, thrombectomy) STEP 1
Thrombosis	Clot forms in place in large or small brain vessels	Atherosclerosis	Reperfusion; prevention
Aneurysm	Dilation of the vessel wall, often at branch points	Asymptomatic until it ruptures → SAH	Clip the stalk
Hemorrhagic stroke	Breakage of a blood vessel	Headache, ↓ consciousness	Control BP, reduce ICP, evacuate hematoma
AVM	Tangle of abnormal vessels linking arteries directly to veins (develop abnormally or change over time) → flow too fast or sluggish → ↓ tissue O₂ ; arterial pressure on weak veins/sinuses → hemorrhage	Most common in the CNS (brain, spine). Early: seizures, headaches, weakness ; may only be a soft "whooshing" bruit	Surgical removal
Dural fistula	AVM where meningeal arteries connect directly to a dural sinus → years of arterial pressure on the sinus	Pulsatile bruit; diagnosed by angiogram	Neuroradiology + surgery (esp. younger patients)

Intracranial hemorrhage types on CT

	INTRAPARENCHYMAL	INTRAVENTRICULAR	SUBARACHNOID	SUBDURAL	EPIDURAL
Location	Inside the brain	Inside the ventricle	Between arachnoid and pia	Between dura and arachnoid	Between dura and skull
Mechanism	High BP, trauma, AVM, tumor	With intraparenchymal or subarachnoid bleeds	Rupture of aneurysm or AVM, or trauma	Trauma	Trauma or after surgery
Source	Arterial or venous	Arterial or venous	Predominantly arterial	Venous (bridging veins)	Arterial (middle meningeal STEP 1)
Shape on CT	Typically rounded	Conforms to ventricles	Tracks along sulci and fissures	Crescent	Lentiform (lens)
Presentation	Acute: sudden headache, nausea, vomiting	Acute: sudden headache, nausea, vomiting	Acute: worst headache of life	May be insidious (worsening headache)	Acute: skull fracture + altered mental status

Imaging

IMAGING	USE
CT	Identifies the type of intracranial hemorrhage
MRI	Rules out a tumor
MRA	Shows arteries: vertebrals, basilar, internal + external carotids, Circle of Willis, cerebrals
Angiogram	Maps vessels; diagnosed the lecture's dural fistula

★ **USMLE high-yield: Embolism** travels, **thrombosis** forms in place (both ischemic). **Aneurysm** = branch-point dilation, silent until rupture → clip. **AVM** = artery-to-vein tangle → seizures, headache, bleed → remove. **Dural fistula** = meningeal artery → sinus. CT shapes: **epidural** = **lentiform** (arterial), **subdural** = **crescent** (bridging veins, insidious), **SAH** = **along sulci** (worst headache, aneurysm). Berry aneurysm risks: ADPKD, Ehlers-Danlos, Marfan **STEP 1**.

Veins & Sinuses · Cerebrospinal Fluid

Veins

GROUP	VEINS	DRAIN TO
External (superficial)	Superficial cerebral veins (superior, middle, inferior); superficial cerebellar veins (superior, inferior)	Nearby dural sinuses (e.g., superior sagittal)
Internal (deep)	Veins of the striatum, thalamus, choroid plexus, hippocampus	Internal cerebral veins → great cerebral vein (of Galen)
Emissary	Connect extracranial (scalp) veins to sinuses	Sinuses; valveless path for infection STEP 1

Dural venous sinuses

Sinuses = spaces between the **outer and inner layers of dura**; **no endothelial cells**; **no valves** (flow not unidirectional). Major: superior sagittal, inferior sagittal, straight, occipital, transverse, superior petrosal, inferior petrosal, sigmoid.



Superior + inferior petrosal sinuses drain from the cavernous sinus toward the transverse/sigmoid sinus and IJV.

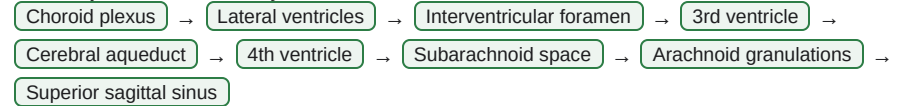
CSF vs plasma

	CSF	PLASMA	IF THE BBB BREAKS
K⁺	2.8 mM	4.5 mM	Alters neuron excitability
Ca²⁺	1.0 mM	2.4 mM	Alters signaling
Protein	34 mg/dL	6500 mg/dL	—
WBCs	0–5	Present	Alters inflammation
RBCs	0	Present	Iron deposition

★ **USMLE high-yield:** Deep veins → internal cerebral veins → **great cerebral vein (Galen)** → straight sinus. Sinuses lie **between dura layers**, with **no endothelium and no valves** → sigmoid sinus → **IJV**. CSF made by the **choroid plexus** (~500 mL/day; ~150 mL present), absorbed by **arachnoid granulations into the superior sagittal sinus**. CSF has less K⁺, Ca²⁺ and protein and almost no cells. Neutrophils = bacterial meningitis; lymphocytes/IgG = MS (oligoclonal bands **STEP 1**); RBCs = SAH.

CSF: production, composition and clearance

CSF: clear, colorless; cushions the brain, gives **buoyancy to ventral arteries**, excretes waste, transports hormones + ions (the brain's "lymphatics"). ~**150 mL** total, **400–500 mL/day** made, pressure **70–180 mm H₂O**. **Choroid plexus:** tufts of specialized **ependymal cells** folded into **villi** in the **lateral, 3rd and 4th ventricles**; filters arterial blood + actively transports some molecules (choroidal artery and vein). **Arachnoid granulations:** arachnoid protrusions through the dura **only in the superior sagittal sinus**; **CSF pressure > venous pressure** → one-way flow into the sinus.



CSF in disease (lumbar puncture)

CSF FINDING	MEANS
Neutrophils, cloudy	Bacterial meningitis
Lymphocytes (IgG)	Active multiple sclerosis
Red blood cells	Subarachnoid hemorrhage
Malignant cells	Metastatic cancer